# Supplementary Results for "Comparing Conditional Co-Participation Across Territories with Survey-Weighted Ising Models"

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*Scope.* These supplementary analyses extend the household descriptions, graphical diagnostics and comparison tests reported in the accompanying manuscript. Their role is to document additional empirical patterns and limits of interpretation; the research question, model definition and core findings are presented independently in the main article. Weighted household summaries and exploratory uncertainty estimates are drawn from the same processed ENAPRES 2025 sample, while predictive diagnostics use the separately executed held-out evaluation.

## 1 Household Composition and Programme Coverage

Seven programme indicators record whether anyone in a household is reported as benefiting from each service. Qali Warma is a school-meal programme, Cuna Más provides early-childhood care and family support, Pensión 65 is non-contributory support for older adults, Juntos is a conditional household transfer, Vaso de Leche and Comedor Popular provide different forms of food assistance, and SIS represents publicly supported health insurance. A household may contribute positive indicators for different members; neither programme dates nor individual-level enrolment matches are observed.

The pooled descriptive prevalence is a weighted household quantity. Table S1 provides its territorial components and illustrates why adjustment is relevant. For example, the much higher rural prevalence of Juntos and Pensión 65 changes the distribution of possible co-participating households. Prevalence differences themselves do not identify variation in conditional Ising interactions.

**Table S1.** Weighted household participation (%) by residence and natural region. Each programme is measured as a separate household-level indicator.

| Territory | *n* | Qali Warma | Cuna Más | Pensión 65 | Juntos | Vaso de Leche | Comedor Popular | SIS |
|---|---|---|---|---|---|---|---|---|
| Urban | 22,269 | 14.6 | 0.9 | 4.6 | 3.0 | 6.0 | 2.2 | 66.4 |
| Rural | 13,915 | 26.7 | 3.8 | 26.2 | 20.6 | 15.3 | 7.7 | 93.0 |
| Coast | 12,919 | 13.0 | 0.7 | 3.3 | 2.2 | 6.4 | 2.9 | 63.6 |
| Highlands | 15,083 | 19.6 | 2.2 | 17.4 | 11.8 | 10.2 | 5.0 | 80.6 |
| Amazon | 8,182 | 28.6 | 2.7 | 12.2 | 12.7 | 9.4 | 1.1 | 85.1 |

The adjusted equations include 23 household regressors with the same meaning in every domain. Demographic composition is represented by the household head's age (per decade), total household size, numbers of children aged 0–11, adolescents aged 12–17 and adults aged 65 or older, the proportion of female members (per 25 percentage points), and whether the head is female. Family organisation is captured by multigenerational structure and presence of a member with a disability. Material circumstances comprise an existing household asset index (per national weighted standard deviation), piped water, sewerage, electricity, internet access and a digital device. Education of the household head is encoded with three contrasts against its first source category; mother tongue has two named-category-number contrasts and one pooled remainder against source category 10. Indigenous identification is coded as yes or missing response, against no. Source category titles for the education and language codes could not be verified in the supplied documentation; substantive labels are therefore not inferred.

All controls enter every programme's conditional equation simultaneously. The asset index is standardised once per training domain/fold; within the original full-sample analysis it is centred and scaled nationally. These are covariate adjustments, not modelled programme eligibility criteria.

### 1.1 Marginal co-participation and socioeconomic composition

Household overlap depends strongly on the pair considered. Table S2 reports a set of contrasts that includes both the largest positive lifts and pairs whose marginal overlap is below independence. A lift above one indicates that two positive indicators co-occur more frequently than suggested by their separate weighted prevalences. It does not imply that the corresponding Ising edge is nonzero after conditioning on the remaining programmes and household characteristics. The Cuna Más–Juntos lift is 8.47, whereas Qali Warma–Pensión 65 has a lift of 0.47; these describe very different household compositions rather than independent administrative decisions.

**Table S2.** Selected marginal weighted co-participation lifts and percentile ranges from 300 grouped resamples. Lift equals one under marginal independence; it is not a conditional Ising parameter.

| Pair | Lift | 2.5th pct. | 97.5th pct. |
|---|---|---|---|
| Qali Warma–Pensión 65 | 0.47 | 0.41 | 0.53 |
| Pensión 65–Juntos | 0.68 | 0.59 | 0.78 |
| Cuna Más–Pensión 65 | 0.56 | 0.37 | 0.77 |
| Qali Warma–Juntos | 3.37 | 3.20 | 3.54 |
| Qali Warma–Vaso de Leche | 2.90 | 2.72 | 3.04 |
| Cuna Más–Juntos | 8.47 | 7.79 | 9.34 |
| Cuna Más–Vaso de Leche | 7.76 | 7.08 | 8.43 |
| Juntos–Vaso de Leche | 4.31 | 3.99 | 4.62 |
| Pensión 65–SIS | 1.36 | 1.34 | 1.38 |
| Pensión 65–Comedor Popular | 2.39 | 2.07 | 2.74 |
| Vaso de Leche–Comedor Popular | 3.09 | 2.65 | 3.64 |

The socioeconomic distribution differs just as markedly (Table S3). In the lowest asset-index quintile, reported Pensión 65 participation is 30.7%, compared with 0.5% in the highest quintile. Internet access also varies by asset position. These gradients motivate adjustment without resolving selection into programme eligibility: the index is an existing derived score, and a household may report beneficiaries of more than one programme.

**Table S3.** Selected household coverage margins by asset-index quintile. Percentage columns are weighted; quintiles are formed from the supplied household asset score.

| Quintile | $n$ | Mean index | Qali Warma | Pensión 65 | Juntos | SIS | Internet |
|---|---|---|---|---|---|---|---|
| 1 | 10,021 | -1.39 | 19.4 | 30.7 | 15.6 | 90.0 | 0.1 |
| 2 | 8,966 | -0.65 | 23.7 | 9.1 | 11.7 | 83.7 | 6.3 |
| 3 | 7,220 | -0.02 | 20.5 | 4.4 | 5.4 | 76.3 | 27.9 |
| 4 | 5,742 | 0.64 | 15.1 | 1.9 | 1.3 | 65.1 | 59.8 |
| 5 | 4,235 | 1.40 | 7.8 | 0.5 | 0.3 | 46.2 | 94.0 |

### 1.2 Age-composition support for joint participation

The opposing age profiles of Qali Warma and Pensión 65 help explain why their low marginal lift need not imply that the programmes are incompatible within a household. Table S4 reports three overlapping age-composition subsets; these are proxies based on household members, not administrative eligibility groups. In 2,211 households containing both younger and older members, weighted Qali Warma prevalence is 31.9%, while Pensión 65 prevalence is 22.8%.

**Table S4.** Programme margins in age-composition subpopulations. Groups overlap by construction; age composition is a proxy, not verified administrative eligibility.

| Age-composition subset | Households | Qali Warma | Pensión 65 | Juntos | SIS |
|---|---|---|---|---|---|
| Children younger than 18 | 17,671 | 37.9 | 3.4 | 14.9 | 80.9 |
| At least one adult 65+ | 11,354 | 6.2 | 26.9 | 1.9 | 68.3 |
| Children and adults 65+ | 2,211 | 31.9 | 22.8 | 9.6 | 82.8 |

An exploratory ridge fit within this mixed-age group returned a Qali Warma–Pensión 65 interaction of 0.263. This subgroup estimate is not part of the territorial test family, and a design-based confidence interval was not available in the executed results. The compositional contrast, rather than this coefficient alone, motivates the main paper's use of household age controls.

## 2 Held-Out Conditional Prediction

The principal manuscript evaluates four specifications over four outer partitions that hold complete conglomerates out of training. For the two Ising models, a separate three-fold grouped validation within each outer training partition chooses the edge penalty. The same conditional loss is computed for each programme, holding the other six observed programme states fixed. Table S5 separates the seven contributions to the pooled gain. This evaluation is not a simulation of previously unknown programme vectors, which would require their joint distribution to be normalised.

**Table S5.** Out-of-cluster conditional log-loss by programme; the last column is the relative reduction for adjusted Ising versus independent covariate models. All estimates use the same household factors.

| Programme | Prevalence | Unadj. Ising | Controls only | Adj. Ising | Gain (%) |
|---|---|---|---|---|---|
| Comedor Popular | 0.1479 | 0.1377 | 0.1377 | 0.1346 | 2.21 |
| Cuna Más | 0.0778 | 0.0578 | 0.0579 | 0.0525 | 9.37 |
| Juntos | 0.2491 | 0.1899 | 0.1603 | 0.1510 | 5.78 |
| Pensión 65 | 0.3096 | 0.2764 | 0.1477 | 0.1353 | 8.38 |
| Qali Warma | 0.4603 | 0.3973 | 0.2920 | 0.2883 | 1.28 |
| SIS | 0.5912 | 0.5216 | 0.4493 | 0.4324 | 3.76 |
| Vaso de Leche | 0.2806 | 0.2315 | 0.2253 | 0.2161 | 4.06 |

The gain is positive for every programme, but its size is uneven. This is consistent with interactions carrying useful conditional information for some outcomes more than others. The pooled difference is 0.008554 log-loss units, a 4.07% reduction relative to the covariates-only benchmark. Its paired 95% percentile range, [0.007805, 0.009315], resamples conglomerates from the fixed out-of-fold losses; it does not account for fitting a different model within each bootstrap replicate.

Table S6 shows the consistency of this result across the four held-out blocks. Preserving distinct source-format conglomerate identifiers rather than casting codes to integers changes the pooled controls-only loss from 0.21002 to 0.21005 and adjusted Ising loss from 0.20147 to 0.20156; the relative gain becomes 4.04%.

This check concerns the predictive ranking only; it does not validate survey strata or uncertainty calculations in the territorial ridge tests.

**Table S6.** Grouped held-out loss by outer fold. Each loss is the weighted mean of seven node-conditional losses.

| Outer fold | Held-out households | Controls only | Adj. Ising | Reduction (%) |
|---|---|---|---|---|
| 1 | 9,046 | 0.21016 | 0.20026 | 4.71 |
| 2 | 9,046 | 0.21206 | 0.20760 | 2.10 |
| 3 | 9,046 | 0.21091 | 0.20141 | 4.50 |
| 4 | 9,046 | 0.20698 | 0.19666 | 4.98 |

The national out-of-fold predictions can be partitioned after estimation by residence and natural region. Table S7 shows the paired comparison in each subset. This is a subgroup evaluation of nationally fitted models, not a separate local cross-validation exercise; the domain rows are overlapping partitions of the same observations and should not be treated as independent replications. The conditional gain is positive in all five domains, largest in the Rural subset.

**Table S7.** Held-out conditional performance of national models evaluated within geographic subsets. The geographic rows share outer-fold fits and are not separately trained local models.

| Evaluation subset | Households | Controls only | Adjusted Ising | Relative gain (%) |
|---|---|---|---|---|
| Urban | 22,269 | 0.1914 | 0.1848 | 3.41 |
| Rural | 13,915 | 0.2768 | 0.2610 | 5.70 |
| Coast | 12,919 | 0.1920 | 0.1860 | 3.11 |
| Highlands | 15,083 | 0.2281 | 0.2161 | 5.26 |
| Amazon | 8,182 | 0.2357 | 0.2257 | 4.24 |

## 3 National Dependence and Selection Stability

The national network is a useful benchmark because it combines the largest domain with the entire range of household characteristics. Table S8 gives every programme pair, including interactions set to zero at the display threshold. The adjusted model is fitted with all 23 controls; the unadjusted comparison uses the same seven programme indicators but no controls. The stability column records the percentage of 75 fixed-penalty cluster resamples in which the adjusted interaction exceeds the display threshold in absolute value. These frequencies should not be interpreted as posterior probabilities or as confidence levels.

**Table S8.** All national programme pairs: regularised coefficients and adjusted-network selection across 75 fixed-penalty resamples. A zero denotes an edge removed by L1 shrinkage.

| Programme pair | Unadjusted $\widehat{\theta}$ | Adjusted $\widehat{\theta}$ | Selected (%) |
|---|---|---|---|
| Qali Warma–Cuna Más | +0.052 | -0.043 | 54.7 |
| Qali Warma–Pensión 65 | -0.985 | +0.000 | 0.0 |
| Qali Warma–Juntos | +1.472 | +0.518 | 100.0 |
| Qali Warma–Vaso de Leche | +1.237 | +0.257 | 100.0 |
| Qali Warma–Comedor Popular | +0.000 | +0.000 | 1.3 |
| Qali Warma–SIS | +1.214 | +0.542 | 100.0 |
| Cuna Más–Pensión 65 | +0.000 | +0.000 | 0.0 |
| Cuna Más–Juntos | +1.685 | +1.086 | 100.0 |
| Cuna Más–Vaso de Leche | +1.793 | +1.036 | 100.0 |
| Cuna Más–Comedor Popular | +0.000 | +0.000 | 0.0 |
| Cuna Más–SIS | +0.096 | +0.000 | 0.0 |
| Pensión 65–Juntos | -0.188 | +0.000 | 0.0 |
| Pensión 65–Vaso de Leche | +0.008 | +0.104 | 76.0 |
| Pensión 65–Comedor Popular | +0.621 | +0.248 | 98.7 |
| Pensión 65–SIS | +2.533 | +2.163 | 100.0 |
| Juntos–Vaso de Leche | +1.039 | +0.707 | 100.0 |
| Juntos–Comedor Popular | +0.378 | +0.020 | 50.7 |
| Juntos–SIS | +2.042 | +1.098 | 100.0 |
| Vaso de Leche–Comedor Popular | +0.858 | +0.833 | 100.0 |
| Vaso de Leche–SIS | +0.934 | +0.476 | 100.0 |
| Comedor Popular–SIS | +0.975 | +0.455 | 100.0 |

The 15 displayed national adjusted edges reduce to 12 under the paper's descriptive criterion of at least 95% resample selection. The three weaker selected edges are Pensión 65–Vaso de Leche (76.0%), Qali Warma–Cuna Más (54.7%) and Juntos–Comedor Popular (50.7%). The last two have small fitted interactions in the adjusted network; their instability contrasts with the national Qali Warma–Juntos and Qali Warma–Vaso de Leche edges, which appear in all 75 resamples.

The complete territorial networks appear in Fig. S1. The main manuscript displays the national and residence subset at larger print scale. Coast, Highlands and Amazon retain substantially different numbers of selected edges under the 1-SE penalty; these should not be read as direct tests of pair-specific territorial differences.

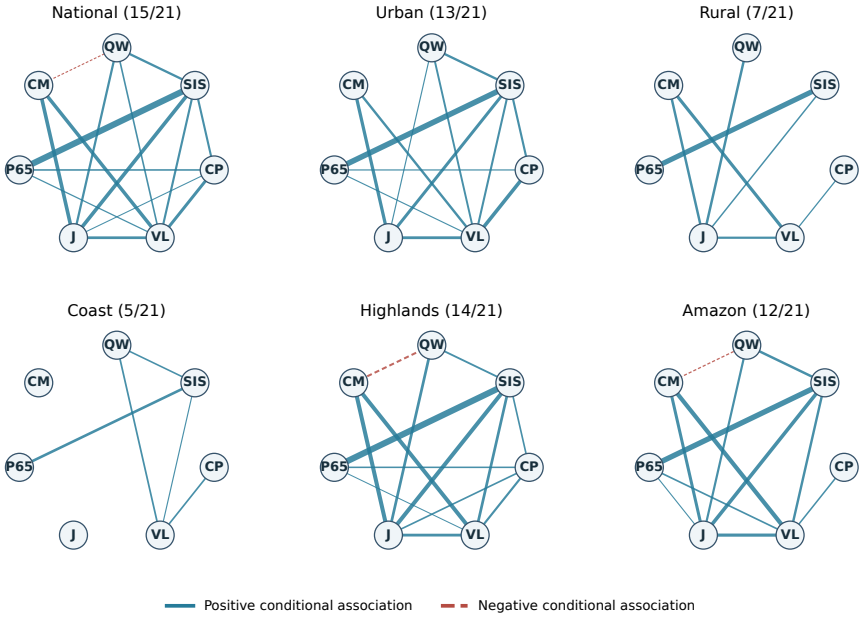



**Fig. S1.** Adjusted seven-node networks for the national sample and five territories. Edges and their thickness encode signed conditional Ising interactions at domain-specific 1-SE penalties, not tests of a territorial difference.

Selection is less uniform outside the pooled network. Table S9 records the number of domain edges that recur in at least 95% of the fixed-penalty resamples. Urban has 13 selected adjusted edges and 10 with this recurrence; Rural has seven selected and six with it. The count of stable edges is a descriptive reproducibility check, not a new hypothesis test. A large coefficient need not be selected with identical frequency across domains because household support and the penalty differ.

**Table S9.** Selection stability in the adjusted national and principal-domain networks. Counts above 95% use fixed-penalty resampling; the percentage denominator is the displayed L1 graph size.

| Domain | Selected edges | $\geq$ 95% | Share (%) | Median selected-edge frequency (%) |
|---|---|---|---|---|
| National | 15 | 12 | 80.0 | 100.0 |
| Urban | 13 | 10 | 76.9 | 100.0 |
| Rural | 7 | 6 | 85.7 | 100.0 |
| Coast | 5 | 3 | 60.0 | 95.0 |
| Highlands | 14 | 13 | 92.9 | 100.0 |
| Amazon | 12 | 8 | 66.7 | 100.0 |

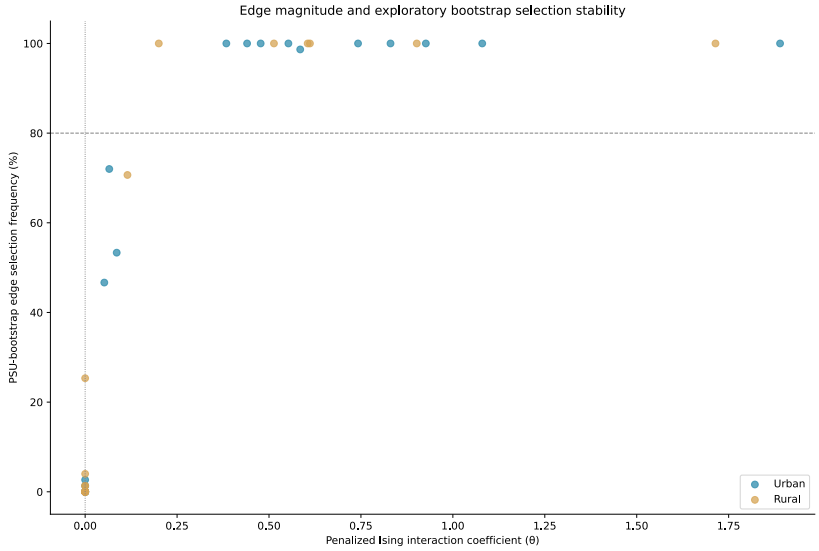



**Fig. S2.** Effect magnitude and bootstrap selection frequency for residence-domain adjusted edges. The dotted horizontal line marks the exploratory 80% visual reference in the source diagnostic; the analysis criterion used in Tables S8 and S9 is 95%, not 80%.

Table S10 identifies the largest reproducibly selected adjusted interactions within each residence and natural-region model. The dominance of Pensión 65–SIS in some domains and the recurring links involving Cuna Más and Juntos are features of conditional co-participation, not evidence of shared administrative operation. All models use the same seven indicators and covariate specification, but have different sample support and penalties; the table ranks interactions only within each domain.

**Table S10.** Largest adjusted interactions among repeatedly selected edges in each principal territory. All edges shown recur in at least 95% of fixed-penalty resamples; sign agreement is calculated conditional on selection.

| Domain | Programme pair | Interaction | Selection (%) | Sign agreement (%) |
|---|---|---|---|---|
| Urban | Pensión 65–SIS | +1.890 | 100.0 | 100.0 |
| | Vaso de Leche–Comedor Popular | +1.080 | 100.0 | 100.0 |
| | Cuna Más–Juntos | +0.927 | 100.0 | 100.0 |
| Rural | Pensión 65–SIS | +1.714 | 100.0 | 100.0 |
| | Cuna Más–Vaso de Leche | +0.902 | 100.0 | 100.0 |
| | Qali Warma–Juntos | +0.611 | 100.0 | 100.0 |
| Coast | Pensión 65–SIS | +0.675 | 100.0 | 100.0 |
| | Qali Warma–Vaso de Leche | +0.310 | 96.7 | 100.0 |
| | Qali Warma–SIS | +0.244 | 95.0 | 100.0 |
| Highlands | Pensión 65–SIS | +2.647 | 100.0 | 100.0 |
| | Cuna Más–Juntos | +1.256 | 100.0 | 100.0 |
| | Juntos–SIS | +1.252 | 100.0 | 100.0 |
| Amazon | Pensión 65–SIS | +1.811 | 100.0 | 100.0 |
| | Cuna Más–Vaso de Leche | +1.342 | 100.0 | 100.0 |
| | Juntos–SIS | +1.080 | 100.0 | 100.0 |

## 4 Adjusted and Reduced-Node Network Specifications

The main analysis estimates a seven-node interaction network in each of the five geographic domains. Removing SIS changes every remaining node-conditional model because the network no longer conditions on health-insurance participation. For that reason, the six-node results are a conditioning-set sensitivity analysis, not a simplified estimate of exactly the same parameters. Table S11 makes the change in graph sizes explicit; the fixed visual layout in Figures S3–S4 permits inspection of which remaining edges are affected.

**Table S11.** Displayed interactions for the three model specifications. The final column refers to a six-node model, not the seven-node adjusted network.

| Domain | Unadj. 7 nodes | Adj. 7 nodes | Adj. 6 nodes | $\lambda_7$ | $\lambda_6$ |
|---|---|---|---|---|---|
| National | 18 | 15 | 10 | 0.00032 | 0.00032 |
| Urban | 15 | 13 | 0 | 0.00032 | 0.00512 |
| Rural | 12 | 7 | 1 | 0.00128 | 0.00512 |
| Coast | 9 | 5 | 0 | 0.00128 | 0.00512 |
| Highlands | 17 | 14 | 0 | 0.00032 | 0.00512 |
| Amazon | 10 | 12 | 4 | 0.00032 | 0.00128 |

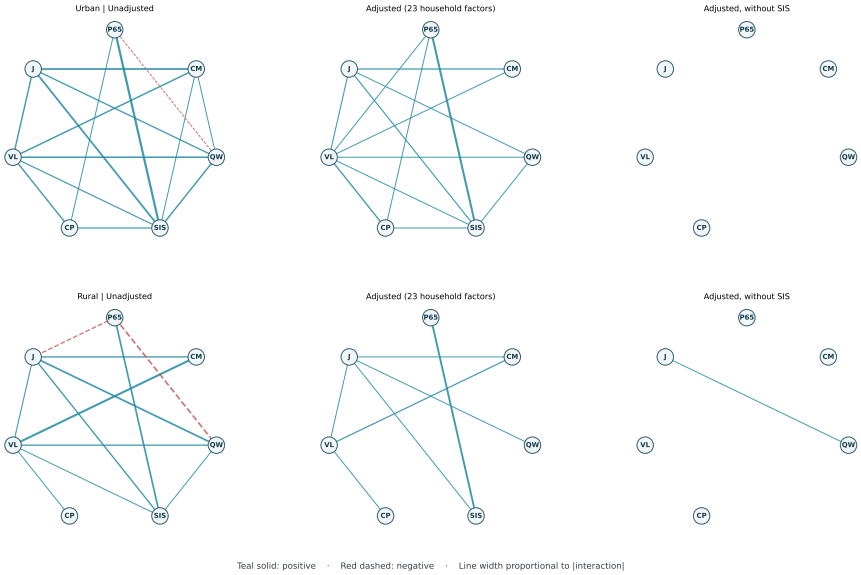



**Fig. S3.** Urban and Rural residence-domain networks under three specifications. QW: Qali Warma; CM: Cuna Más; P65: Pensión 65; J: Juntos; VL: Vaso de Leche; CP: Comedor Popular; SIS: health insurance. Each panel uses its own 1-SE-selected penalty and the same node positions.

In Urban households, covariate adjustment reduces the displayed graph from 15 to 13 edges; removing SIS then selects none under the six-node 1-SE penalty. In Rural households, the corresponding counts are 12, seven and one. These changes are large on the graph representation, but they combine omitted-variable conditioning with fresh penalty selection. They cannot be interpreted as a causal or statistical effect of SIS enrolment on other programmes.

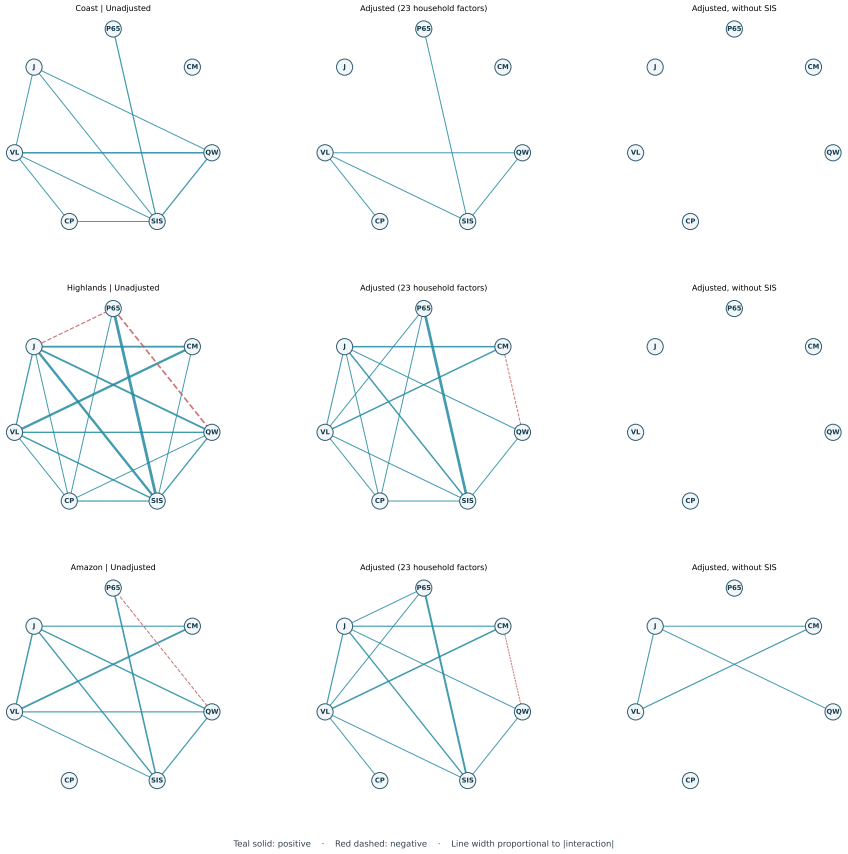



**Fig. S4.** Coast, Highlands and Amazon networks under the unadjusted, 23-covariate adjusted, and adjusted-without-SIS specifications. All edges are conditional associations from domain-specific penalised fits; the graphs do not test between-region differences.

The Highlands retains more displayed adjusted links than the Coast, even though both regions contain all seven programme indicators. As discussed in the article, domain-specific edge counts are affected by the tuning criterion and the distribution of covariates. The ridge contrasts are needed to assess pair-specific changes on a common seven-node parameter scale.

## 5 Why Marginal Overlap and L1 Edge Changes Are Not Territorial Tests

Marginal weighted log-odds-ratio contrasts are inexpensive to calculate, but they concern a different estimand from the conditional Ising interaction. Similarly, a change in the presence of a selected L1 edge is a discontinuous property of regularisation; it need not accompany a large ridge parameter contrast. Table S12 quantifies these discrepancies separately for the four geographic comparisons.

The overlap counts cannot be interpreted as accuracy scores because no method supplies ground-truth labels for heterogeneous programme pairs.

**Table S12.** Two descriptive alternatives to ridge contrasts. "Sign discordant" counts opposite signs of marginal and ridge coefficient differences; "both" means agreement of BH selection for marginal comparisons or co-occurrence of an L1 edge toggle and a ridge-selected contrast.

| Comparison | Opp. signs | Marg. BH | Ridge BH | Both BH | L1 flips | Both |
|---|---|---|---|---|---|---|
| Rural–Urban | 7 | 10 | 7 | 1 | 6 | 1 |
| Highlands–Coast | 8 | 11 | 8 | 4 | 11 | 6 |
| Amazon–Coast | 6 | 9 | 6 | 3 | 9 | 3 |
| Amazon–Highlands | 11 | 5 | 3 | 1 | 4 | 2 |
| Total (84 pairs) | 32 | 35 | 24 | 9 | 30 | 12 |

In particular, Qali Warma–Cuna Más is not selected in the Urban or Rural L1 graph, yet the corresponding ridge difference is $-0.383$ (BH $q = 0.0039$). This illustrates why edge toggles are not substitutes for tests on a common parameter. Marginal and adjusted contrasts also depend on different conditioning sets; a marginal-only significant result must not be described as a false positive relative to the adjusted analysis.

Table S13 selects pairs that illustrate distinct ways in which marginal comparisons, the L1 selection display and ridge parameter tests can disagree. Qali Warma–Cuna Más does not toggle between the Urban and Rural L1 networks and its marginal difference is approximately zero, yet the adjusted ridge contrast is selected after BH correction. Conversely, Qali Warma–Comedor Popular has a marginally selected residence difference, with no corresponding adjusted ridge evidence. Neither estimator supplies ground truth for the other because they address different estimands.

**Table S13.** Contrasting examples for marginal log-odds, adjusted ridge differences and L1 edge presence. The two adjusted $q$-columns refer to distinct estimands; an L1 toggle is not a hypothesis test.

| Territories | Pair | Marg. $\Delta$ | Ridge $\Delta$ | Marg. $q$ | Ridge $q$ | L1 toggle |
|---|---|---|---|---|---|---|
| Rural–Urban | Qali Warma–Cuna Más | +0.008 | -0.383 | 0.971 | 0.004 | No |
| Rural–Urban | Qali Warma–Juntos | +0.281 | +0.542 | 0.061 | 0.000 | No |
| Rural–Urban | Qali Warma–Vaso de Leche | -0.163 | -0.346 | 0.158 | 0.009 | Yes |
| Rural–Urban | Qali Warma–Comedor Popular | -0.474 | -0.091 | 0.026 | 0.721 | No |
| Highlands–Coast | Cuna Más–Juntos | -0.315 | +0.448 | 0.452 | 0.024 | Yes |
| Highlands–Coast | Pensión 65–SIS | +0.807 | +0.692 | 0.180 | 0.000 | No |
| Amazon–Coast | Comedor Popular–SIS | +1.540 | -0.361 | 0.000 | 0.002 | No |
| Amazon–Highlands | Pensión 65–SIS | -0.741 | -0.539 | 0.150 | 0.000 | No |

## 6 Territorial Interaction Differences

The main manuscript presents representative ridge contrasts together with the full 84-pair colour map. Tables S14–S15 supply the additional numerical results that survive BH adjustment in their respective test families. The uncertainties are approximate: territorial fits are separate and weighted, 150 group resamples are used per domain, and official ENAPRES strata are absent. The minimum unweighted $2 \times 2$ cell across compared territories is displayed because rare joint cells can make the normal approximation unreliable even when $q$ is small.

**Table S14.** BH-selected ridge differences: Rural–Urban and Highlands–Coast. All quantities are on the conditional log-odds scale; $m$ is the minimum unweighted cell, with † denoting $m < 10$.

| Contrast | Programme pair | $\Delta$ | Boot. SE | BH $q$ | $m$ |
|---|---|---|---|---|---|
| Rural/Urban | Qali Warma–Cuna Más | -0.383 | 0.115 | 0.00387 | 149 |
| | Qali Warma–Pensión 65 | -0.262 | 0.099 | 0.025 | 151 |
| | Qali Warma–Juntos | +0.542 | 0.131 | 0.000276 | 464 |
| | Qali Warma–Vaso de Leche | -0.346 | 0.115 | 0.00901 | 733 |
| | Cuna Más–Vaso de Leche | +0.609 | 0.120 | 7.46e-06 | 140 |
| | Pensión 65–SIS | +0.452 | 0.116 | 0.000507 | 20 |
| | Juntos–SIS | +0.271 | 0.066 | 0.000276 | 2† |
| Highlands/ | | | | | |
| Coast | Qali Warma–Cuna Más | -0.327 | 0.122 | 0.0336 | 73 |
| | Qali Warma–Juntos | +0.621 | 0.137 | 9.1e-05 | 220 |
| | Qali Warma–Vaso de Leche | -0.607 | 0.131 | 7.9e-05 | 466 |
| | Cuna Más–Juntos | +0.448 | 0.158 | 0.024 | 56 |
| | Cuna Más–Vaso de Leche | +0.600 | 0.152 | 0.000833 | 67 |
| | Pensión 65–SIS | +0.692 | 0.108 | 1.03e-08 | 19 |
| | Juntos–SIS | +0.274 | 0.064 | 0.000212 | 2† |
| | Vaso de Leche–Comedor Popular | -0.434 | 0.174 | 0.0489 | 135 |

**Table S15.** BH-selected ridge differences: Amazon–Coast and Amazon–Highlands. All quantities are on the conditional log-odds scale; $m$ is the minimum unweighted cell, with † denoting $m < 10$.

| Contrast | Programme pair | $\Delta$ | Boot. SE | BH $q$ | $m$ |
|---|---|---|---|---|---|
| Amazon/Coast | Qali Warma–Vaso de Leche | -0.540 | 0.159 | 0.00471 | 336 |
| | Cuna Más–Vaso de Leche | +0.682 | 0.176 | 0.000987 | 67 |
| | Pensión 65–Comedor Popular | -0.404 | 0.141 | 0.024 | 22 |
| | Juntos–SIS | +0.209 | 0.080 | 0.0378 | 1† |
| | Vaso de Leche–Comedor Popular | -0.528 | 0.194 | 0.0321 | 33 |
| | Comedor Popular–SIS | -0.361 | 0.098 | 0.00177 | 0† |
| Amazon/ | | | | | |
| Highlands | Pensión 65–Juntos | +0.345 | 0.114 | 0.0157 | 88 |
| | Pensión 65–Comedor Popular | -0.223 | 0.090 | 0.0496 | 22 |
| | Pensión 65–SIS | -0.539 | 0.108 | 1.93e-05 | 12 |

The major contrasts are not a simple monotonic shift across programmes. The Rural–Urban ridge coefficient is higher for Qali Warma–Juntos and Cuna Más–Vaso de Leche, but lower for Qali Warma–Vaso de Leche. Highlands–Coast follows the former two directions for Qali Warma–Juntos and Qali Warma–Vaso

de Leche. Pensión 65–SIS is highest in the Highlands among the three natural regions. These interpretations refer to conditional co-participation parameters, not changes in eligibility or administrative coordination. Four selected contrasts involve sparse observed cells and should be treated particularly cautiously.

The four BH-selected results with sparse joint cells require particular restraint. Table S16 shows the joint-positive counts and the smallest of the four cell counts across the compared samples. A minimum cell count of zero can occur despite a nonzero ridge coefficient because the model is regularised and includes the other programme states and covariates. Approximate Wald statistics in such settings are not a substitute for a method tailored to separation or the exact survey design. We report these tests but do not treat them as decisive evidence of population heterogeneity.

**Table S16.** BH-selected contrasts with at least one sparse $2 \times 2$ cell. The two $n_{11}$ columns count joint-positive households; $m$ is the smallest of all four cells across the two domains.

| Contrast | Pair | $n_{11}$ ref. | $n_{11}$ comp. | $m$ | BH $q$ |
|---|---|---|---|---|---|
| Rural–Urban | Juntos–SIS | 959 | 2392 | 2 | 0.00028 |
| Highlands–Coast | Juntos–SIS | 451 | 1806 | 2 | 0.00021 |
| Amazon–Coast | Juntos–SIS | 451 | 1094 | 1 | 0.03782 |
| Amazon–Coast | Comedor Popular–SIS | 502 | 114 | 0 | 0.00177 |

Figure S5 presents the numerical values of all 84 territorial differences, including pairs that do not pass the BH screen. Its four columns share the same pair ordering as the compact visual summary in the main article; the annotation is descriptive and does not remove the sparse-cell caution.

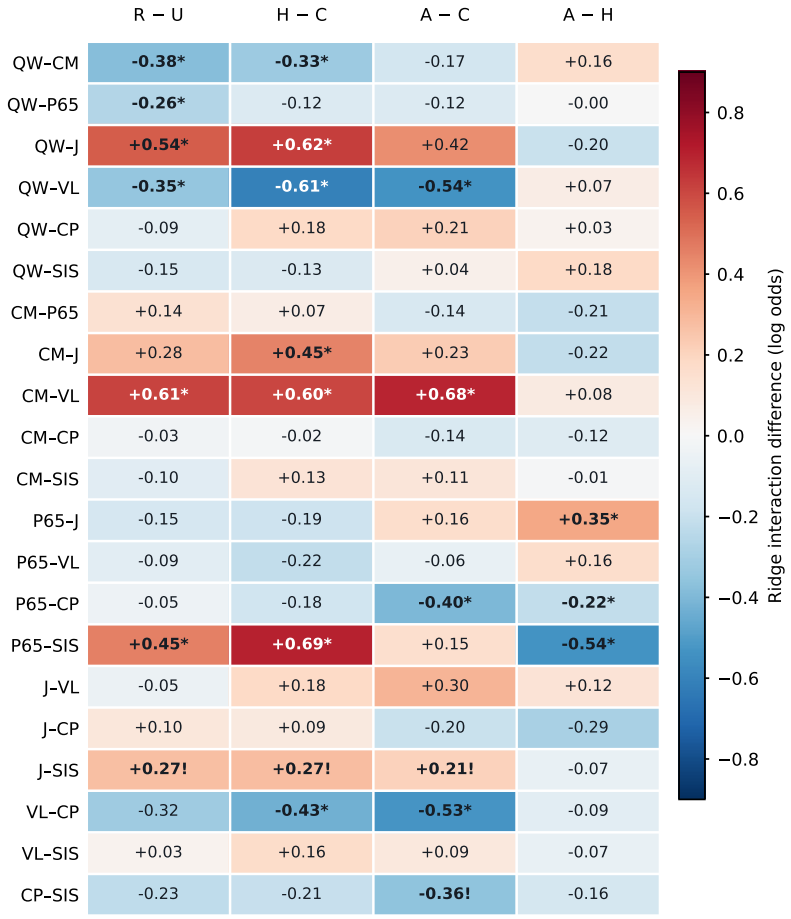



**Fig. S5.** Complete pairwise ridge interaction differences in their fitted conditional log-odds scale. Asterisks mark BH-selected comparisons and exclamation marks flag sparse-cell contrasts, which are reported in Tables S14–S15.

## 7 Household Characteristics: Additional Conditional Associations

Table S17 reports selected conditional odds ratios from the national adjusted Ising fit. For a continuous regressor, the reported odds ratio compares the stated increment with other covariates and other programme indicators held fixed. For binary indicators, it compares presence with absence. Bootstrap percentile bounds hold the network penalty fixed; they do not represent causal effects of household composition or economic position.

**Table S17.** Selected programme-specific conditional odds ratios from the adjusted national network. The 95% bootstrap ranges are descriptive; other programme states and all remaining household regressors are held fixed.

| Programme | Household characteristic and unit | OR | Lower | Upper |
|---|---|---|---|---|
| Qali Warma | Children aged 0–11 (per child) | 3.528 | 3.248 | 3.788 |
| Qali Warma | Head age (per decade) | 0.854 | 0.826 | 0.880 |
| Cuna Más | Children aged 0–11 (per child) | 1.654 | 1.522 | 1.800 |
| Cuna Más | Household size (per person) | 1.328 | 1.250 | 1.422 |
| Pensión 65 | Adults aged 65+ (per person) | 4.621 | 4.203 | 5.096 |
| Pensión 65 | Household asset index (per 1 SD) | 0.339 | 0.314 | 0.375 |
| Juntos | Household asset index (per 1 SD) | 0.391 | 0.366 | 0.427 |
| Juntos | Internet at home (yes vs. no) | 0.569 | 0.506 | 0.639 |
| Vaso de Leche | Children aged 0–11 (per child) | 1.998 | 1.843 | 2.176 |
| Vaso de Leche | Head age (per decade) | 0.849 | 0.796 | 0.897 |
| Comedor Popular | Household asset index (per 1 SD) | 0.628 | 0.562 | 0.700 |
| Comedor Popular | Piped water access (yes vs. no) | 1.254 | 1.121 | 1.371 |
| SIS | Household size (per person) | 1.960 | 1.869 | 2.073 |
| SIS | Household asset index (per 1 SD) | 0.517 | 0.474 | 0.556 |

Age composition and the asset index show sizeable but programme-specific conditional associations. Another child aged 0–11 is associated with substantially higher conditional participation odds in Qali Warma and Vaso de Leche, while older-adult composition has the strongest association with Pensión 65. The asset index is inversely associated with several income- or food-related programmes. None of these conditional odds ratios is a direct targeting measure; programme eligibility and the reported indicators need not coincide.

Further conditional odds ratios are collected in Table S18. The within-domain associations of children with Qali Warma and of older adults with Pensión 65 remain positive, but their magnitude differs across the five domains. Since these odds ratios come from separate penalised fits, simple differences between table entries should not be given the inferential interpretation of the ridge interaction contrasts. Their purpose is to show how the adjusted equations differ in composition while preserving the same set of explanatory variables.

**Table S18.** Selected programme-conditional household odds ratios by domain, each from separately fitted adjusted Ising models. These descriptive contrasts have not undergone formal territorial tests.

| Programme | Household factor | Urban | Rural | Coast | Highlands | Amazon |
|---|---|---|---|---|---|---|
| Qali Warma | Children aged 0–11 (per child) | 3.70 | 3.22 | 3.59 | 3.99 | 3.12 |
| Cuna Más | Children aged 0–11 (per child) | 1.66 | 1.72 | 1.73 | 1.84 | 1.53 |
| Pensión 65 | Adults aged 65+ (per person) | 3.34 | 8.07 | 2.95 | 6.09 | 6.65 |
| Juntos | Household asset index (per 1 SD) | 0.46 | 0.51 | 0.45 | 0.37 | 0.63 |
| Vaso de Leche | Children aged 0–11 (per child) | 1.93 | 2.19 | 1.90 | 2.42 | 1.89 |
| Comedor Popular | Household asset index (per 1 SD) | 0.62 | 0.92 | 0.53 | 0.55 | 0.91 |
| SIS | Household size (per person) | 1.95 | 2.00 | 1.85 | 2.23 | 2.36 |

## 8 Departmental Scope of Network Estimation

Department-level estimates were not included in the main heterogeneity tests: some programme indicators lack the support needed for a seven-node fit at this smaller scale. Figure S6 reports the number of estimable programme nodes for all departments; Table S19 identifies the departments with exclusions. The omitted programmes vary by department, so comparing the resulting graphs as if they represented the same node set would be misleading.

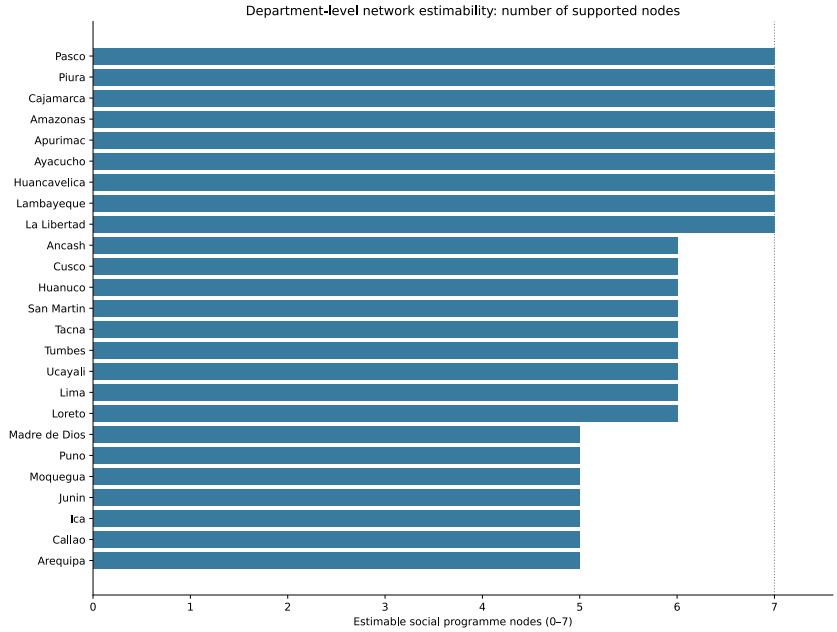



**Fig. S6.** Number of programme indicators with sufficient observed support for estimation within each department (maximum seven). This figure concerns model estimability, not programme effectiveness or household coverage.

**Table S19.** Departments for which fewer than seven programme indicators met the source support rule. Other departments retained all seven; omitted nodes reflect insufficient sample support, not necessarily zero participation.

| Department | Households | Groups | Nodes | Omitted programme indicators |
|---|---|---|---|---|
| Arequipa | 1,474 | 174 | 5 | Cuna Más, Comedor Popular |
| Callao | 1,342 | 241 | 5 | Cuna Más, Juntos |
| Ica | 1,142 | 166 | 5 | Cuna Más, Juntos |
| Junin | 1,463 | 164 | 5 | Cuna Más, Comedor Popular |
| Madre de Dios | 1,091 | 156 | 5 | Cuna Más, Comedor Popular |
| Moquegua | 1,347 | 147 | 5 | Cuna Más, Juntos |
| Puno | 1,458 | 175 | 5 | Cuna Más, Comedor Popular |
| Ancash | 1,377 | 172 | 6 | Cuna Más |
| Cusco | 1,427 | 170 | 6 | Cuna Más |
| Huanuco | 1,459 | 161 | 6 | Comedor Popular |
| Lima | 2,997 | 520 | 6 | Cuna Más |
| Loreto | 1,469 | 158 | 6 | Comedor Popular |
| San Martin | 1,685 | 199 | 6 | Comedor Popular |
| Tacna | 1,048 | 129 | 6 | Cuna Más |
| Tumbes | 1,157 | 117 | 6 | Cuna Más |
| Ucayali | 1,263 | 153 | 6 | Comedor Popular |

The reduced-node department models are best read as descriptive local profiles. Geographic heterogeneity tests in the main analysis were therefore restricted to Urban–Rural and natural-region contrasts, which retain the same seven indicators

and adjustment specification. For departmental inference, a new analysis with verified design strata, adequate programme support and comparable parameter spaces would be required.

## 9 Penalty Sensitivity and Scope of Inference

An L1-selected graph is a diagnostic representation, not the basis of the pairwise territorial tests. Table S20 reports the effect of penalty choice on graph size across the national sample and five principal domains. A lower selected $\lambda$ need not imply that a domain has a substantively denser dependency network; penalty choice reflects the validation distribution and effective sample support.

**Table S20.** Adjusted seven-node network selection across major domains. The last columns show the displayed number of interactions under different tuning rules.

| Domain | $n$ | 1-SE $\lambda$ | Min-CV $\lambda$ | 1-SE edges | Min-CV edges | Common-$\lambda$ edges |
|---|---|---|---|---|---|---|
| National | 36,184 | 0.00032 | 5e-06 | 15 | 20 | 15 |
| Urban | 22,269 | 0.00032 | 5e-06 | 13 | 21 | 13 |
| Rural | 13,915 | 0.00128 | 5e-06 | 7 | 21 | 15 |
| Coast | 12,919 | 0.00128 | 2e-05 | 5 | 19 | 12 |
| Highlands | 15,083 | 0.00032 | 2e-05 | 14 | 20 | 14 |
| Amazon | 8,182 | 0.00032 | 5e-06 | 12 | 21 | 12 |

Four of the six main-domain minima are at the smallest tested penalty, so the location of the actual optimum is not bracketed in those domains. The 1-SE rule nevertheless selects appreciably sparser graphs. With the health-insurance indicator excluded, each six-node graph also changes its conditioning set; the resulting edges are not directly comparable with those from the seven-node model. The main paper's ridge contrasts remain on the common seven-node adjusted scale.

The strongest restrictions on population inference are the unavailable official sampling strata, uncertainty in the household expansion factor for these derived outcomes and low joint-cell counts for four of the 24 BH-selected territorial differences. Original skip patterns and programme histories are not retained in the processed household file. We therefore interpret the estimated networks and associated contrasts as an exploratory description of reported household co-participation, and the bootstrap intervals as approximate uncertainty under the implemented grouped resampling rather than official design-based confidence intervals.